From Social Dilemmas to Artificial Societies: Measuring Baseline, Steerability, and Institutional Behavior in LLM Agents

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Abstract

Large language models are increasingly deployed as interactive agents, but their social behavior is still often summarized with a single headline number: how cooperative a model appears on one benchmark under one prompt. That summary is too coarse. In social environments, measured behavior can reflect at least four separable factors: a model's baseline policy under minimally directive prompts, its susceptibility to prompt framing, its sensitivity to canonical benchmark presentation, and its adaptation to institutions such as scarcity or public reputation. We present an evaluation framework organized explicitly around that decomposition and apply it to repeated Prisoner's Dilemma, Chicken, Stag Hunt, and small artificial-society settings.

The current audited result bundle supports five main conclusions. First, game structure strongly shapes baseline behavior: cooperation rises from Prisoner's Dilemma (`0.4722 / 0.5463`) to Chicken (`0.9074 / 0.7870`) to Stag Hunt (`0.9722 / 0.9907`) on the stable triplet cohort, and the same qualitative ordering appears in a same-day replication cohort. Second, neutral wording is not behaviorally innocuous. In pooled Prisoner's Dilemma runs, the literal `minimal-neutral` wording averages `0.2639 / 0.2917` cooperation, whereas two more abstract neutral paraphrases average `0.5833 / 0.6667`; the compact and institutional paraphrases yield identical action traces on all 12 matched pairings, so we collapse them into a single abstract-neutral family and obtain an exact paired randomization p-value of `0.03125`. Third, prompt framing can move the same audited Prisoner's Dilemma setup from universal defection to universal cooperation. Fourth, benchmark presentation materially changes measured behavior, but not monotonically across games: unnamed framing raises cooperation in Prisoner's Dilemma while lowering it in Chicken and Stag Hunt. Fifth, institutional prompts alter visible social structure more reliably than they improve collective survival: in scarcity, `task-only` achieves the best final survival, while public reputation equalizes survival without erasing differences in trade and alliances.

Taken together, the results argue that behavioral alignment work should not ask whether models are "cooperative" in the abstract. It should ask which behavior is being measured, under which prompt, in which game, and under what institutional frame.

1. 1. Introduction

Alignment is often discussed as if it were a property that can be read off from static safety benchmarks or from whether a model refuses an obviously harmful request. Many real deployments, however, are strategic and social. Models negotiate, coordinate, bargain, advise, compete, and act in institutional settings where short-horizon incentives can conflict with collective outcomes. In those settings the relevant question is not just whether a model can follow instructions. It is how the model behaves when the environment makes cooperation valuable, risky, or manipulable.

Repeated games and artificial societies offer a natural way to study that question because they expose several distinct failure modes that are easy to conflate. A model may appear cooperative on a canonical benchmark only because it recognizes the task label. Another model may look cooperative only under a prompt that explicitly asks it to be prosocial. A third may produce visible trade and alliance behavior in a society simulation while still degrading the group's survival prospects. Treating all of those outcomes as one scalar "cooperation" score obscures the real empirical structure.

This paper therefore frames LLM social evaluation as a decomposition problem. We separate four targets that are often mixed together in prior work:

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1. baseline behavior under minimally directive prompts
2. steerability under cooperative or competitive framing
3. sensitivity to canonical benchmark presentation
4. adaptation to institutions such as scarcity and public reputation

That decomposition matters methodologically. If neutral paraphrases shift behavior, then a paper that reports one default Prisoner's Dilemma cooperation rate is overstating the precision of its own baseline. If canonical benchmark labels move behavior, then benchmark familiarity is a confound rather than a detail. If prompt framing can swing the same setup from universal defection to universal cooperation, then prompt obedience and default policy should not be treated as the same construct. If institutions preserve survival without preserving the same social dynamics, then visible prosociality and resilient collective outcomes should be analyzed separately.

The contribution of the current paper is not the generic observation that LLMs can cooperate or compete. That space is already crowded. The contribution is a more careful measurement framework paired with an audited empirical bundle that shows why the decomposition is necessary. We focus on three repeated games where the evidence is deepest, then connect them to two small multi-agent institutions. The result is a paper about measurement, not just a paper about levels.

The most important manuscript-facing contributions are:

1. A repeated-game evaluation design that distinguishes baseline behavior, prompt steerability, and benchmark recognizability instead of collapsing them.
2. An audited Prisoner's Dilemma baseline result showing that even neutral wording must be treated as a family of defensible baselines rather than as a single point estimate.
3. A cross-game comparison showing that both baseline cooperation and prompt sensitivity depend strongly on strategic structure.
4. A society/reputation extension showing that institutions can preserve survival while still producing sharply different social organization.

The current paper is strongest as a methodological and empirical submission on behavioral evaluation. It is not a claim that we have solved the full micro-to- macro alignment problem. It is a claim that current practice often measures the wrong thing too coarsely, and that a more careful decomposition already changes the substantive interpretation of LLM social behavior.

2. 2. Related Work

The most relevant prior work falls into four clusters: classical-game studies of LLM strategy, work on altruism and human social preferences, multi-agent simulation and institutional design, and broader multi-agent evaluation frameworks.

In classical games, the closest predecessors are Lorè and Heydari (2024), Akata et al. (2025), and Fontana et al. (2024). Lorè and Heydari are especially important because they already show that context and game structure both matter for strategic behavior, making prompt sensitivity a methodological issue rather than a nuisance variable. Akata et al. extend repeated-game analysis across broader task families and anchor the literature on finitely repeated strategic interaction. Fontana et al. focus specifically on iterated Prisoner's Dilemma behavior and provide a comparison point for cooperation rates and policy style. Relative to this line of work, our goal is not to rediscover that prompts matter. It is to turn prompt sensitivity, benchmark recognition, and baseline instability into separate design axes within one evaluation bundle.

Work on altruism, fairness, and human-comparison benchmarks provides a second anchor. Schmidt et al. (2024) show that GPT-3.5's advice in altruistic settings is sensitive to reciprocal concerns. Capraro et al. (2025) provide a benchmark for predicting how humans balance self-interest against others' welfare. These studies are important because they discourage naive claims that LLM outputs straightforwardly reveal stable moral traits. At the same time, they are not primarily about repeated strategic interaction among LLM agents, and they do not ask whether benchmark names or neutral wording alter measured behavior in the games themselves.

On the multi-agent side, Park et al. (2023) introduced generative agents as a general architecture for socially situated simulations. Sreedhar et al. (2025) and Piao et al. (2025) are closer to our institutional direction: they show that multi-agent LLM systems can produce meaningful social patterns and policy- relevant behavior in larger environments. These papers motivate our move from two-player games to scarcity and public-reputation societies, but they do not center the bridge between micro-level strategic measurement and macro-level institutional outcomes. Our society experiments are deliberately narrower and more game-theoretic: the aim is not to simulate all social life, but to test how prompt-conditioned behavioral tendencies survive or fail under institutional pressure.

Finally, benchmark-oriented frameworks such as ALYMPICS (Mao et al., 2024) and MultiAgentBench (Zhu et al., 2025) provide useful context for evaluation infrastructure. They broaden the space of collaboration

and competition tasks for LLM agents, but their primary contribution is not the specific separation of baseline behavior, framing obedience, benchmark recognition, and institutional mediation. That separation is the cleanest novelty gap for the present work.

The paper therefore positions itself at the intersection of these literatures. It uses repeated games as the most controlled setting for baseline, steerability, and recognition effects, then uses small institutions to test whether those prompt-conditioned behaviors survive contact with collective constraints. The resulting claim is narrower than "LLMs can act socially" but more useful for evaluation design.

3. 3. Experimental Design

3.1. 3.1 Measurement targets

The framework treats social evaluation as a set of related but distinct measurement problems.

Baseline measurement asks what a model does under minimally directive prompts. This is the closest analogue to a default social policy, but even here we do not assume there is one canonical neutral phrasing. Prompt susceptibility asks how much behavior moves when the system and framing layers explicitly encourage cooperative or competitive interpretation. Benchmark-recognition measurement asks whether canonical task labels activate different policies than unnamed or disguised isomorphs. Institutional measurement asks whether these micro-level behavioral tendencies survive under social environments with resource pressure and public reputation.

3.2. 3.2 Model cohort and task families

The audited Part 1 bundle uses a stable three-model cohort:

- `cerebras:llama3.1-8b`
- `nvidia:deepseek-ai/deepseek-v3.2`
- `nvidia:moonshotai/kimi-k2-instruct-0905`

A same-day replication cohort substitutes Qwen for Kimi:

- `cerebras:llama3.1-8b`
- `nvidia:deepseek-ai/deepseek-v3.2`
- `cerebras:qwen-3-235b-a22b-instruct-2507`

The cohort is intentionally pragmatic rather than aspirational. Models were selected because they were both accessible and action-ready during the April 7 to April 8, 2026 runs. That makes the panel dated and imperfect, but it also makes the reported results reproducible. We do not present the panel as a representative sample of closed frontier models, and we return to that limitation explicitly in Section 6.

The repeated-game portion centers on Prisoner's Dilemma, Chicken, and Stag Hunt because those three games already separate conflict, anti-coordination, and coordination incentives. The institutional portion uses a scarcity society and a public-reputation variant of the same world.

3.3. 3.3 Repeated-game protocol

Each repeated-game experiment uses self-play plus all cross-model pairings, yielding six matched pairings per cohort. The audited experiments cited in this paper use six rounds per trial at temperature `0.0`, with one trial per pairing-condition combination. Baseline experiments compare three neutral prompt variants. Susceptibility experiments compare `minimal-neutral`, `cooperative`, and `competitive` prompts. Benchmark experiments compare canonical labels against unnamed and, for Prisoner's Dilemma, resource-disguise presentations.

This design is small-n at the trial level, but it supports paired inference: the same model pairing can be compared across prompt conditions, which is more informative than treating each condition as a fully independent sample. That paired structure is especially important for the baseline-instability result, where the strongest evidence comes from within-pairing shifts rather than from between-experiment aggregate means.

3.4. 3.4 Metrics and statistical reporting

The paper reports cooperation and average payoff in the repeated games, plus survival, trade volume, inequality, commons health, and alliance count in the institutions. We report agent-position metrics because asymmetries sometimes matter, but for inferential tests we collapse to a per-trial mean across the two positions.

We use two forms of uncertainty reporting. First, result summaries and figures show bootstrap confidence intervals on trial means. These intervals are descriptive, not inferential. Overlap or non-overlap between two bootstrap intervals is not treated as a formal hypothesis test. Second, when the design admits a clean matched comparison, we use an exact two-sided sign-flip randomization test on paired trial-level mean cooperation. This is the inferential statistic used for the manuscript's focal baseline and steerability contrasts.

3.5. 3.5 Audit protections

Two audit protections matter for interpretation. First, the current result bundle preserves the full message stack through `messages_sent`, rather than only the final game prompt. That is necessary because the effective input contains sys-

tem, framing, persona, and history layers. Second, summary tooling de-duplicates interrupted retries and prefers the latest completed JSON result for each logical experiment name. This avoids stale partial JSONL files being double-counted in manuscript-facing summaries and figures.

4. 4. Results

4.1. 4.1 Strategic environment dominates simple cooperation labels

The stable triplet cohort produces a clear cross-game ordering. Aggregate cooperation is lowest in Prisoner's Dilemma, higher in Chicken, and highest in Stag Hunt:

The same qualitative ordering appears in the same-day replication cohort: Prisoner's Dilemma at `0.4815 / 0.5370`, Chicken at `0.6852 / 0.6389`, and Stag Hunt at `0.9722 / 0.9907`. The levels move, especially in Chicken, but the ranking is stable. This matters because it argues against a simple model- wide label such as ”cooperative” or ”competitive.” The strategic environment is doing substantial explanatory work.

4.2. 4.2 Neutral baselines are a family, not a point estimate

The strongest baseline-instability result is in Prisoner's Dilemma. Pooling two cohorts yields the following descriptive summary:

The key point is not merely that the means differ. It is that the two more abstract neutral paraphrases produced identical action traces on all 12 matched pairings across both cohorts while still eliciting different raw natural- language reasoning. That makes the result easier to interpret. This is not a file-selection or condition-label bug. The compact and institutional prompts are distinct inputs, but for this specific game and model panel they collapse behaviorally to the same policy.

We therefore treat them as an abstract-neutral family for inference rather than as two independent discoveries. On matched pairings, the abstract-neutral family exceeds `minimal-neutral` by `+0.3472` mean per-trial cooperation, with an exact paired sign-flip p-value of `0.03125`. Put differently: the literal neutral wording and the abstract neutral wording families are not estimating the same baseline policy on this audited Prisoner's Dilemma bundle.

This is a stronger and more reviewer-proof claim than ”three neutral prompts all differ significantly.” We do not have evidence for that stronger claim, and the compact/institutional collapse argues against it. The defensible conclusion is narrower and still important: there is no single prompt-free baseline here. There is at least a two-family neutral baseline problem, and papers that report one default cooperation estimate without robustness checks risk overstating precision.

4.3. 4.3 Prompt framing is highly steerable, but not uniformly so

Prompt susceptibility is strongest in Prisoner's Dilemma. On the stable cohort, `competitive` yields `0.0000 / 0.0000` cooperation, `cooperative` yields `1.0000 / 1.0000`, and `minimal-neutral` remains much lower at `0.2500 / 0.3056`. The exact paired sign-flip comparison between competitive and cooperative framing is `p = 0.03125` on the six matched pairings because every pairing moves in the same direction.

Chicken and Stag Hunt are also steerable, but they retain far more residual cooperation under competitive framing:

The correct reading is therefore not simply that prompt framing matters. It is that steerability itself is game-dependent. In some environments framing can almost fully determine observed social behavior; in others it changes behavior without erasing the underlying strategic affordances of the game.

4.4. 4.4 Benchmark presentation matters, but the current evidence is still small-sample

Benchmark presentation materially changes the measured Prisoner's Dilemma result. The canonical presentation yields `0.2500 / 0.3056` cooperation, the resource-disguise presentation yields `0.5833 / 0.6667`, and the unnamed presentation yields `0.7500 / 0.8056`. The mean per-trial cooperation increase from canonical to unnamed is `+0.5000`.

However, this is where inferential restraint matters. The benchmark track uses only six matched pairings on the stable cohort. An exact paired sign-flip test for canonical versus unnamed Prisoner's Dilemma gives `p = 0.0625`, while the canonical versus resource contrast gives `p = 0.25`. Those p-values do not erase the very large descriptive effect sizes, but they do tell us that the benchmark-recognition result should be framed as strong directional evidence with limited power rather than as a precision estimate.

The cross-game pattern is still informative. Chicken and Stag Hunt move in the opposite direction from Prisoner's Dilemma: unnamed framing lowers cooperation relative to canonical. That reversal is theoretically important because it argues against a naive story in which removing benchmark labels simply reveals some deeper underlying cooperation tendency. Presentation is interacting with the strategic schema a model appears to activate for each game.

Table 1. Repaired cross-game baseline on the stable triplet cohort.

Game	Trials	Cooperation A	Cooperation B	Average Payoff A	Average Payoff B
Prisoner's Dilemma	18	0.4722	0.5463	2.2500	1.8796
Chicken	18	0.9074	0.7870	2.5093	2.9907
Stag Hunt	18	0.9722	0.9907	3.9630	3.9074

Table 2. Pooled repaired Prisoner's Dilemma neutral-variant replication across two cohorts.

Prompt Variant	Pooled Trials	Cooperation A	Cooperation B	Average Payoff A	Average Payoff B
minimal-neutral	12	0.2639	0.2917	1.6806	1.5417
minimal-neutral-compact	12	0.5833	0.6667	2.5000	2.0833
minimal-neutral-institutional	12	0.5833	0.6667	2.5000	2.0833

4.5. 4.5 Institutions separate visible prosociality from collective survival

The two institutional settings yield a cleaner macro-level result than the earlier pilot: visible sociality and resilient collective outcomes do not move in lockstep.

In the scarcity society, `task-only` achieves the best final survival (`1.0000`) while both `cooperative` and `competitive` fall to `0.8889`. Yet those two lower-survival conditions do not look the same. `Cooperative` produces the most trade and alliance activity, along with the highest inequality and stronger commons recovery. `Competitive` remains alliance-free and much less unequal. Public reputation changes the picture again: all three prompt families preserve `1.0000` final survival, but `cooperative` still produces much more trade and alliance structure than either `task-only` or `competitive`.

This is exactly the kind of dissociation that a one-number cooperation metric misses. A prompt can make an environment look more visibly prosocial without improving survival, and an institution can equalize survival without equalizing the rest of social organization.

4.6. 4.6 Summary of inferential support

For clarity, Table 1 summarizes the exact paired tests for the manuscript's three focal repeated-game claims.

The table makes the paper's intended evidential posture explicit. The baseline- instability and prompt-steerability results now have matched-pair inferential support on the audited bundle. The benchmark-presentation result is still a large descriptive shift, but it remains underpowered in the current stable cohort and should be written that way.

5. 5. Discussion

Three broader lessons follow from these results.

First, LLM social behavior is not well described by a single latent trait. The same model panel can look exploitative in one game, cooperative in another, hyper-responsive to prompt framing in one setting, and relatively insensitive in another. That finding is conceptually important because much of the public discussion around model alignment still talks as if models can be assigned a stable social label. Our data do not support that style of summary.

Second, neutral-baseline measurement deserves to be treated as a methodological robustness problem. In many empirical literatures, a neutral prompt is treated as a default control condition whose wording is not itself of scientific interest. In social LLM evaluation that assumption is unsafe. The audited Prisoner's Dilemma result shows that a literal neutral wording and a more abstract neutral wording family can induce materially different behavior even when neither prompt is overtly prosocial or adversarial. That does not mean all neutral paraphrases are equally unstable, and indeed our compact and institutional phrasings collapsed behaviorally. But it does mean that a single "baseline" estimate should not be presented without prompt-robustness context.

Third, institutions should not be evaluated only through their effect on headline survival or cooperation rates. The scarcity and reputation results show that institutions can preserve or equalize survival while still producing very different trade, alliance, and inequality regimes. From an alignment perspective, that matters because visible prosocial activity may not be a good proxy for resilient collective outcomes. The right institutional question is not only whether agents act socially, but whether the induced social structure is compatible with the objective the institution is supposed to protect.

The paper's framing also suggests a useful way to interpret benchmark effects. Canonical task names should be treated as part of the prompt distribution, not as a transparent window onto some model's "true" social preference. If a model changes behavior when the benchmark name is removed, that is itself an important behavioral fact. The correct response is not to decide that either the canonical or unnamed condition is uniquely real. It is to recognize that

Table 3. Repaired prompt susceptibility across the three core games.

Game	Prompt Variant	Trials	Cooperation A	Cooperation B
Prisoner's Dilemma	competitive	6	0.0000	0.0000
Prisoner's Dilemma	cooperative	6	1.0000	1.0000
Chicken	competitive	6	0.6389	0.3333
Chicken	cooperative	6	1.0000	1.0000
Stag Hunt	competitive	6	0.4167	0.3056
Stag Hunt	cooperative	6	1.0000	1.0000

Table 4. Repaired prompt susceptibility across the three core games.

Contrast	Matched Trials	Mean Trial Cooperation A	Mean Trial Cooperation B	Mean Difference (B - A)
PD minimal-neutral vs abstract-neutral family	12	0.2778	0.6250	0.3472
PD competitive vs cooperative	6	0.0000	1.0000	1.0000
PD canonical vs unnamed	6	0.2778	0.7778	0.5000

recognizability is a first-class axis of evaluation.

6. 6. Limitations

The current submission has six important limitations.

First, the repeated-game cohort is small and pragmatic. It is multi-model and multi-provider, but it is not a frontier panel. The paper should therefore not claim that the reported levels are representative of the strongest closed models.

Second, the repeated-game trial count is still limited. Even with paired comparisons, six matched pairings per cohort is not enough to make every large effect statistically decisive. We therefore distinguish between descriptive effect size and inferential support rather than pretending they are the same thing.

Third, the benchmark track is especially underpowered. The Prisoner's Dilemma presentation effect is descriptively large, but the exact paired p-value for canonical versus unnamed is `0.0625` on the stable cohort. That is directionally valuable evidence, not a license to write the result as a settled law.

Fourth, the institutional battery is still small at three repetitions per prompt family in each institution. The social-structure patterns are already interesting, but the correct tone is ”current evidence” rather than ”institutional universals.”

Fifth, the current baseline result depends on a specific neutral-family decomposition. Compact and institutional prompts are distinct inputs, yet they collapse to identical Prisoner's Dilemma action traces on the pooled matched trials. We respond by collapsing them into an abstract-neutral family for the inferential claim, but future work should test a wider bank of neutral phrasings and larger model panels before turning this into a generalized taxonomy.

Sixth, provider availability is temporally unstable. The cohort reported here is tied to the models that were accessible and action-ready during the April 2026 execution window. Future reruns may require substitutions, and those substitutions should be treated as empirical changes rather than invisible maintenance.

These limitations do not erase the paper's main contribution. They bound it. The strongest claim is not that the current experiments settle LLM social behavior. It is that a careful evaluation framework already changes what we can say responsibly with the data we have.

7. 7. Reproducibility And Release Considerations

The project is structured so that the paper can be regenerated from versioned artifacts rather than from hand-maintained tables. Prompts live as text files, experiments are config-driven, raw results are preserved in JSON or JSONL artifacts, and summary/figure generation is handled by dedicated scripts. That layout makes the paper auditable in a way that many benchmark reports are not: the manuscript can point directly to the result directories and regeneration commands that produced its tables and figures.

The release posture should still be careful. These results are behavioral and contextual. They do not show that a model is intrinsically altruistic, selfish, or morally aligned in any deep sense. They show that behavior in social environments is shaped by game structure, prompt framing, benchmark recognition, and institutional rules. Open release is useful because it lets others inspect those conditions directly, but the accompanying narrative should avoid anthropomorphic overclaiming.

The statistical reporting policy should also remain explicit in the released bundle. Bootstrap confidence intervals are descriptive. Matched exact paired tests are inferential where the design supports them. Older artifacts that do not pre-

serve the full message stack should not be treated as the manuscript record when newer audited artifacts exist.

8. 8. Conclusion

The central lesson of the paper is methodological. Before asking whether LLMs are cooperative, altruistic, or aligned in the abstract, we need to ask which behavior is being measured, under which prompt, in which game, and under what institutional frame. Once those factors are separated, the empirical picture is less tidy but more useful: baseline behavior is game-dependent, neutral wording is not innocuous, prompt steerability is large but uneven, benchmark recognition is real, and institutions can preserve survival without preserving the same social structure. That is a better foundation for social-behavior evaluation than any single canonical cooperation score.

This appendix collects the manuscript-adjacent details needed to reproduce the current paper bundle from the repository.

A. A. Artifact Map

Primary manuscript artifacts:

- `paper/MANUSCRIPT.md`
- `paper/RESULTS_TABLES.md`
- `paper/FIGURE_CAPTIONS.md`
- `paper/REFERENCES.md`
- `paper/ARTIFACT_INDEX.md`

Primary result summaries:

- `results/paper_ready_baseline_triplet/interim_summary.md`
- `results/paper_ready_replications/interim_summary.md`
- `results/paper_ready_benchmark_triplet/interim_summary.md`
- `results/paper_ready_susceptibility_triplet/interim_summary.md`
- `results/paper_ready_society_triplet/interim_summary.md`
- `results/paper_ready_reputation_triplet/interim_summary.md`

Primary manuscript-facing figures:

- `paper/figures/repaired_pd_replications/baseline_prompt_variants_cooperation.png`
- `paper/figures/triplet_live/baseline_prompt_variants_cooperation.png`
- `paper/figures/benchmark_live/benchmark_presentations_cooperation.png`
- `paper/figures/susceptibility_live/susceptibility_prompt_variants_cooperation.png`
- `paper/figures/society_reputation_live/society_reputation_final_survival.png`
- `paper/figures/society_reputation_live/society_reputation_trade_volume.png`
- `paper/figures/society_reputation_live/society_reputation_alliance_count.png`

Primary statistical appendix outputs:

- `paper/tables/paired_statistical_tests.csv`
- `paper/tables/paired_statistical_tests.md`

B. B. Experimental Settings

B.1. B.1 Part 1 repeated-game cohorts

Stable triplet cohort:

- `cerebras:llama3.1-8b`
- `nvidia:deepseek-ai/deepseek-v3.2`
- `nvidia:moonshotai/kimi-k2-instruct-0905`

Same-day replication cohort:

- `cerebras:llama3.1-8b`
- `nvidia:deepseek-ai/deepseek-v3.2`
- `cerebras:qwen-3-235b-a22b-instruct-2507`

Protocol highlights:

- self-play plus all cross-model pairings
- six repeated rounds per trial in the cited Part 1 artifact bundle
- temperature `0.0`
- one trial per pairing-condition combination
- full `messages_sent` prompt-stack logging in the audited artifacts

B.2. B.2 Part 2 scarcity society

Population:

- two agents per model from the stable triplet cohort

World settings:

- initial public resources `36`

- maximum public resources `48`
- regeneration rate `0.2`
- initial agent resources `6`
- gather amount `3`
- steal amount `2`
- survival cost `1`
- reproduction threshold `15`
- offspring start resources `4`
- maximum agents `12`

Institutional settings:

- private messages enabled
- stealing enabled
- unmonitored agents enabled
- unmonitored fraction `0.2`
- trade-offer TTL `3`

B.3. B.3 Part 3 public reputation society

Part 3 uses the same world configuration as Part 2, with:

- public reputation enabled
- reputation decay `0.95`
- anonymous ratings disabled
- rating range `1` to `5`
- no unmonitored agents

C. C. Metrics And Statistical Reporting

C.1. C.1 Repeated-game metrics

- cooperation by agent position
- average payoff by agent position
- pooled per-trial mean cooperation for paired inference
- benchmark-presentation deltas
- prompt-susceptibility deltas

C.2. C.2 Society metrics

- `survival_rate`
- `final_survival_rate`
- `average_trade_volume`
- `average_gini`
- `commons_health`
- `alliance_count`
- `extinction_event`
- `commons_depletion_rate`

C.3. C.3 Reporting convention

The paper uses two different uncertainty summaries.

Descriptive summaries:

- pooled means
- 95% bootstrap confidence intervals on trial means

Inferential summaries:

- exact two-sided sign-flip randomization tests on matched trial-level mean cooperation when the design supports a paired comparison

This distinction matters. Confidence-interval overlap is not treated as a hypothesis test in the manuscript. The focal Prisoner's Dilemma baseline and steerability claims rely on the paired exact tests instead.

C.4. C.4 Neutral-family handling

In pooled Prisoner's Dilemma baseline runs, the compact and institutional neutral prompts are distinct inputs but produce identical action traces on all 12 matched pairings across the two cohorts. The paper therefore treats them as an `abstract-neutral` family for the inferential baseline claim and reports the compact-versus-institutional comparison separately as a null contrast.

D. D. Reproduction Commands

All commands below were run from the repository root.

Regenerate manuscript-facing summaries:

```
.venv/bin/python scripts/paper_summary.py \
  results/paper_ready_baseline_triplet \
  results/paper_ready_replications \
  results/paper_ready_benchmark_triplet \
```

```
  results/
      paper_ready_susceptibility_triplet \
  results/paper_ready_society_triplet \
  results/paper_ready_reputation_triplet \
  --markdown paper/summary.md \
  --csv paper/summary.csv
```

Regenerate paired statistical tests:

```
.venv/bin/python scripts/paper_stats.py
```

Regenerate manuscript-facing figures:

```
.venv/bin/python scripts/paper_figures.py \
  results/paper_ready_baseline_triplet/paper
      -baseline-prisoners_dilemma-20260407
      T173331Z.json \
  results/paper_ready_replications/paper-
      baseline-prisoners_dilemma-20260407
      T172819Z.json \
  --output-dir paper/figures/
      repaired_pd_replications

.venv/bin/python scripts/paper_figures.py \
  results/paper_ready_baseline_triplet \
  --output-dir paper/figures/triplet_live

.venv/bin/python scripts/paper_figures.py \
  results/paper_ready_benchmark_triplet \
  --output-dir paper/figures/benchmark_live

.venv/bin/python scripts/paper_figures.py \
  results/
      paper_ready_susceptibility_triplet \
  --output-dir paper/figures/
      susceptibility_live

.venv/bin/python scripts/paper_figures.py \
  results/paper_ready_society_triplet \
  results/paper_ready_reputation_triplet \
  --output-dir paper/figures/
      society_reputation_live
```

Build the anonymous ICML-style submission PDF:

```
.venv/bin/python scripts/
    build_icml_submission.py
```

Run paper-related regression coverage:

```
.venv/bin/python -m pytest -q \
  tests/test_paper_summary.py \
  tests/test_paper_figures.py \
  tests/test_paper_batch.py \
  tests/test_paper_stats.py \
  tests/test_build_icml_submission.py \
  tests/test_experiments.py
```

E. E. Threats To Validity

The current paper bundle improves over earlier pilot artifacts, but several validity constraints still matter.

First, the result panel is date-specific. The accessible and action-ready model cohort for the April 2026 runs should be treated as an empirical object, not as a timeless standard benchmark.

Second, the benchmark and susceptibility contrasts remain small-sample. Large descriptive shifts can coexist with exact p-values that remain above common thresholds because only six matched pairings are available in the stable cohort.

Third, the institutional battery is still modest in scale at three repetitions per prompt family. This supports cautious claims about survival-sociality dissociations, not sweeping claims about institutional law.

Fourth, the current prompt audit uncovered a nuisance prompt-construction issue in the framing layer: a literal player identifier placeholder is preserved in some framing prompts. Because the issue is shared across compared neutral conditions and the logged outputs show condition-specific raw responses, we do not treat it as an explanation for the compact/institutional collapse. It should nevertheless be corrected in future reruns.

F. F. Ethics And Release

This project should be released as a behavioral benchmark and artifact bundle, not as evidence that models possess stable moral traits. Terms such as `cooperative`, `competitive`, or `aligned` should be interpreted as conditioned behavioral descriptors under specific prompts, games, and institutions.

Prompt files, configs, summaries, and logs are appropriate to release because they enable auditing and replication. The main ethical risk is misinterpretation: narrow behavioral findings can easily be overread as claims about intention, value, or character. The release bundle should therefore pair open artifacts with explicit statements about scope, sample size, dated provider availability, and the difference between visible prosociality and collective resilience.

G. Selected Figures

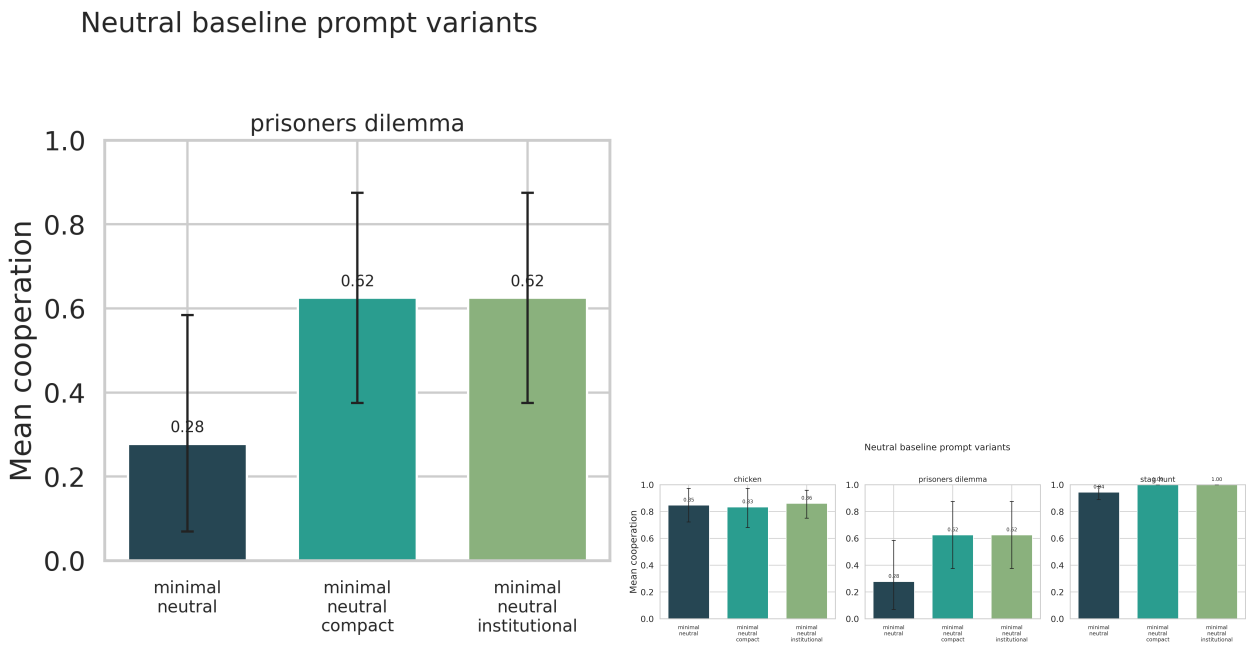

Figure 1. Neutral-baseline robustness in Prisoner's Dilemma (left) and cross-game baseline comparison (right). The pooled PD figure makes the central baseline-instability claim legible at full size, while the cross-game figure shows that strategic environment dominates simple model-wide cooperation labels.

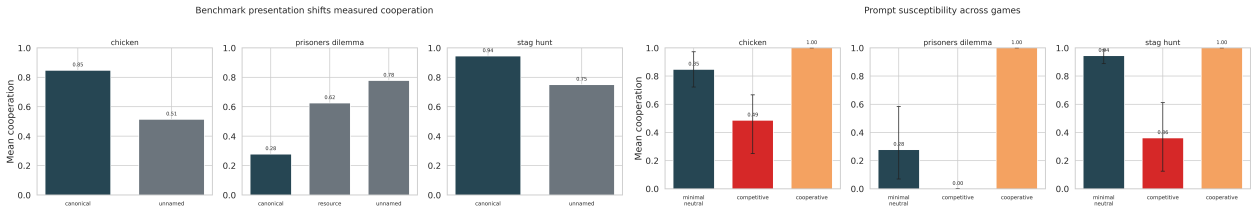

Figure 2. Benchmark-presentation effects (left) and prompt susceptibility (right). Both results are shown at a larger scale than the earlier three-across layout so the game-level differences remain readable in the submission PDF.

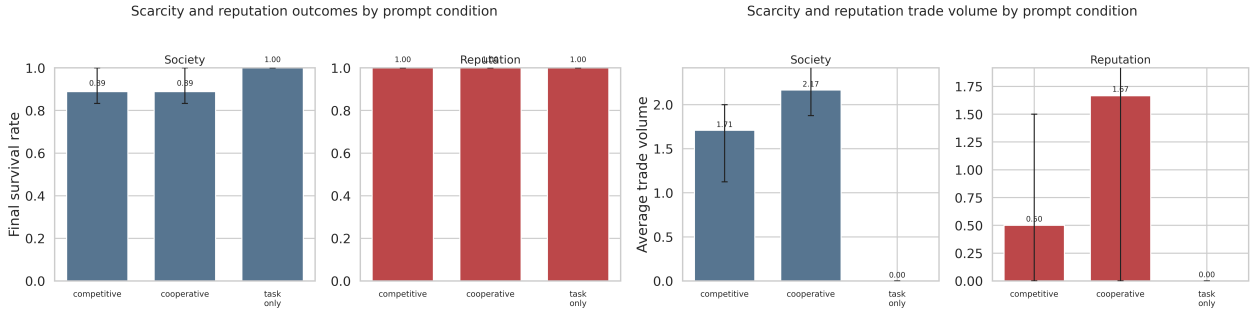

Figure 3. Institutional outcomes under scarcity and public reputation. Final survival is shown on the left and trade volume on the right. Public reputation equalizes survival while preserving large prompt-conditioned differences in social activity.

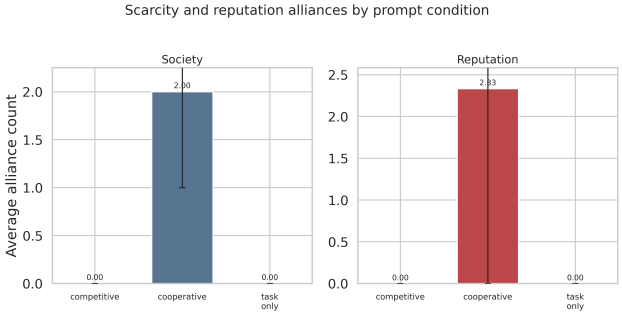

Figure 4. Alliance formation under scarcity and public reputation. Cooperative prompting produces visible alliance structure even when final survival converges under public reputation.

References

Core references currently used in the manuscript draft.

1. Nunzio Lore and Babak Heydari. *Strategic behavior of large language models and the role of game structure versus contextual framing*. Scientific Reports, 2024. URL: https://www.nature.com/articles/s41598-024-69032-z

2. Elif Akata et al. *Playing repeated games with large language models*. Nature Human Behaviour, 2025. URL: https://www.nature.com/articles/s41562-025-02172-y

3. Nicoló Fontana, Francesco Pierri, and Luca Maria Aiello. *Nicer Than Humans: How do Large Language Models Behave in the Prisoner's Dilemma?* arXiv, 2024. URL: https://arxiv.org/abs/2406.13605

4. Manuel Schmidt et al. *GPT-3.5 altruistic advice is sensitive to reciprocal concerns but not to strategic risk*. Scientific Reports, 2024. URL: https://www.nature.com/articles/s41598-024-73306-x

5. Valerio Capraro, Roberto Di Paolo, and Veronica Pizziol. *A publicly available benchmark for assessing large language models' ability to predict how humans balance self-interest and the interest of others*. Scientific Reports, 2025. URL: https://www.nature.com/articles/s41598-025-01715-7

6. Karthik Sreedhar, Alice Cai, Jenny Ma, Jeffrey V. Nickerson, and Lydia B. Chilton. *Simulating Cooperative Prosocial Behavior with Multi-Agent LLMs: Evidence and Mechanisms for AI Agents to Inform Policy Decisions*. arXiv, 2025. URL: https://arxiv.org/abs/2502.12504

7. Jinghua Piao et al. *AgentSociety: Large-Scale Simulation of LLM-Driven Generative Agents Advances Understanding of Human Behaviors and Society*. arXiv, 2025. URL: https://arxiv.org/abs/2502.08691

8. Joon Sung Park, Joseph C. O'Brien, Carrie J. Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. *Generative Agents: Interactive Simulacra of Human Behavior*. arXiv, 2023. URL: https://arxiv.org/abs/2304.03442

9. Shaoguang Mao et al. *ALYMPICS: LLM Agents Meet Game Theory -- Exploring Strategic Decision-Making with AI Agents*. arXiv, 2024. URL: https://arxiv.org/abs/2311.03220

10. Kunlun Zhu et al. *MultiAgentBench: Evaluating the Collaboration and Competition of LLM agents*. In Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), 2025. URL: https://aclanthology.org/2025.acl-long.421/